DPS integration: cause and effect

Review the conditions, actions and interfaces needed to protect equipment and coordinate operation.

> Working draft. We still need owner agreement and tests on the installed paths.
> 46 rules · 16 sources · 5 decisions

Conditions → actions

A blank cell means this row assigns no action there. Read each condition with its action, sources and open decisions.

Released, draft and design describe the source. Inference = proposed rule. Open = missing rule. Conflict = sources differ.

OFF Stop or remove power from the named load. **INHIBIT** Withdraw the permit or inhibit the named equipment. **BLOCK** Deny the conflicting operating request. **LOCAL** Facility or equipment controller acts locally; see the rule. **ALARM** Annunciate the stated alarm; no detector trip is assigned by this cell.
MONITOR Monitor or report the stated status. **TAG** Publish activity and scope for data tagging. **NOTIFY** Send the named event/status; this does not stop DAQ. **REVIEW** Action or priority is unresolved. **CONTINUE** Keep the named local functions running within the stated limits.

Rule	Condition	Evidence	Drift HV	FC termination	Top CRP bias	BDE power	TDE power	PDS bias / power	Lights / lasers	CI / calibration	Rack AC	Facility plant	SC / operator	DAQ information	Sources
Facility															
FAC-01	Fire alarm applies to the detector area	Released	OFF	OFF	REVIEW										S01, S03
DAQ-01	DAQ barracks or Ross Dry emergency AC shutdown	Released									OFF				S01
ODH-01	Oxygen-deficiency alarm or personnel-safety condition	Open	REVIEW									LOCAL	REVIEW		S01, S12, S15
FAC-02	Oro Hondo / SDSTA ventilation status unavailable or not satisfactory	Open										REVIEW	REVIEW		S12, S14
BMS-01	Fire/smoke override for the named ventilation equipment	Design										LOCAL			S13
BMS-02	Supply-fan high discharge or low suction pressure switch trips	Design										OFF			S13
BMS-03	Sump-pump common alarm	Design										ALARM			S13
BMS-04	BMS UPS alarm	Design										ALARM			S13
BMS-05	Active condenser basin reaches the specified low-low or high-high level	Design										OFF			S13
BMS-06	Required cooling service is lost while its detector load is powered	Inference				REVIEW					REVIEW				S13, S10, S11
BMS-07	Rack or room temperature exceeds its monitoring limit	Inference									REVIEW		ALARM		S10, S11
BMS-08	BMS operator station or controller network fails	Design										CONTINUE			S13
Rack infrastructure															
RACK-01	Smoke detected in a protected rack	Released									OFF		MONITOR		S01, S11
Cryogenics / HV															
CRP-01	LAr falls below the top CRP composite frame while top CRP bias is on	Released			OFF										S01
HV-01	LAr approaches the HV feedthrough shielding rim while drift HV is on	Released	OFF												S01, S03
HV-02	Pressure relief valve opens while drift HV is on	Released	OFF												S01, S03
HV-05	Another condition could cause shallow LAr boiling while drift HV is on	Released	OFF												S01, S03
HV-03	Sudden unexpected cathode HV current increase	Released	OFF												S01, S03
HV-04	Cathode HV supply shuts down	Released		OFF											S01, S03

Rule	Condition	Evidence	Drift HV	FC termination	Top CRP bias	BDE power	TDE power	PDS bias / power	Lights / lasers	CI / calibration	Rack AC	Facility plant	SC / operator	DAQ information	Sources
HV-06	Excess cathode HV current / streamer detected	Released						NOTIFY						NOTIFY	S01, S03
CI-01	Unexpected temperature change or possible recirculation problem	Released											MONITOR		S01
CI-02	LAr level, temperature-readout state and PrM flash-lamp state	Released											MONITOR		S01
BDE															
DPS-C001	WIEC fan cooling is inadequate while BDE electronics are powered	Inference				INHIBIT									S04, S06, S16
BDE-02	Flange heat is inadequate or flange approaches dew point	Inference				REVIEW									S04, S16
BDE-03	WIB temperature crosses its listed SC alarm bound	Draft											ALARM		S05
BDE-04	PTC/WIB protection data or control path is lost	Open				REVIEW									S04, S02, S06
PDS / optical															
Q-001	Purity-monitor flash lamp is active	Conflict						REVIEW						REVIEW	S01, S07, S12, S08
O-001	Camera illumination is requested while the relevant PDS state is active	Conflict						REVIEW	REVIEW				NOTIFY		S07, S03, S08
O-002	Ionisation or alignment laser conflicts with a PDS operation	Draft							BLOCK				NOTIFY		S08
Q-002	Declared LCM light calibration is running	Draft												TAG	S08
C-001	PoF supply is unavailable or reports its own fault	Draft						REVIEW	LOCAL						S08, S07
P-011-I	PDS supply current is outside its specified range	Released						OFF							S01, S07, S08
P-011-V	PDS supply voltage is outside its specified range	Released						OFF							S01, S07, S08
P-012-S	PDS power supply reports overtemperature	Released						OFF							S01, S07, S08
P-012-R	PDS readout module reports overtemperature	Released						OFF							S01, S07, S08
P-012-L	Light Calibration Module reports overtemperature	Released						OFF							S01, S07, S08
TDE / other															
TDE-01	TDE crate/board local fault	Draft					LOCAL								S09
TDE-02	TDE condition requiring an external DPS inhibit	Open					REVIEW								S09
CRYO-01	Cryogenic instrumentation itself needs equipment protection	Open								REVIEW					S01
PNS-01	Pulsed-neutron calibration condition needs exclusion or protection	Open								REVIEW					S08, S12
GND-01	Grounding/impedance-monitoring alarm	Open											REVIEW		S10
SC / DPS / DAQ															
SC-01	SC server, display or normal monitoring link fails	Inference											ALARM		S04, S01, S12
SC-02	Operator requests reset or re-enable after a trip	Inference											REVIEW		S01, S11
DPS-C006	Protection controller, input module or its power supply fails	Open	REVIEW	REVIEW	REVIEW	REVIEW	REVIEW	REVIEW			REVIEW				S14, S01
DAQ-02	SC detects an approaching equipment limit before a protection trip	Draft											ALARM	NOTIFY	S11
DAQ-03	Routine SC command would disturb equipment in an active DAQ run	Draft											BLOCK	MONITOR	S11

Decisions needed

D01 · Who executes software protection?

TTO 33801 p16 places software interlocks in SC. The proposed SC–DPS boundary assigns protection to DPS/local controllers. Dedicated BDE paths do not settle ownership for every function.

Agree: SC and DPS must list each protection function, its executor, review authority and failure behavior. Confirm whether Fermilab takes full DPS accountability.

With: SC and DPS leaders; technical coordination; subsystem owners

S12 PDF p16, §4.2 (TTO requirement)
S04 PDF p7, §5 (Stated requirement)

D02 · PrM flash lamp: bias off or tagging?

The general baseline, EDMS PDS attachment and TTO require protecting SiPMs with bias off/special configuration. The newer working PDS matrix allows acquisition with tagging.

Agree: PDS and PrM owners must resolve optical risk and affected scope before an operating rule is chosen. A newer draft is not approval to remove an interlock.

With: PDS; PrM/CI; DPS; SC; DAQ

S01 PDF p12 (Stated requirement)
S07 PDF p7 (Working draft)
S12 PDF pp24,28,40 (TTO requirement)
S08 Q-001 (Proposed rule)

D03 · Camera LEDs and PDS: which request has priority?

The old PDS text turns bias off. HVS text describes blocking LEDs while PDS is active. The working PDS draft denies conflicts but leaves priority/state details open. The old text also repeats PrM in the camera sentence.

Agree: Agree permitted combinations, request priority, proven-off feedback and settling time; correct the copied wording.

With: Camera/HVS owner; PDS; DPS; SC

S07 PDF p7 (Working draft)
S03 PDF pp6,8 (Working draft)
S08 O-001 (Proposed rule)

D04 · Which BMS revision and points describe the installed plant?

The online-linked baseline is Bulletin 11 in DocDB 30544 v2. The supplied Bulletin 13 has no established online identity here. Neither establishes installed I/O or a commissioned detector interface.

Agree: Obtain the current approved/commissioned sequences, I/O list, panel drawings, point schedule and as-built mapping. Compare them with the cabinet survey.

With: ISD/BSI; facility controls owner; SDSTA; I&I; DPS

S13 Section 230923 (Design requirement)

D05 · Alarm bounds are not automatically trip settings

The BDE worksheet includes an 80 °C upper alarm bound; HVS gives twice normal current for over 1 s as an example; SC–I&I gives 20 °C as a design intake condition. None is an approved DPS setting in this matrix.

Agree: Equipment owners must supply the damage limits and margin, delay, calibration, measurement validity and affected load for each protection rule.

With: Each equipment owner; DPS; SC

S05 WIBFEMB row 5 (Working draft)
S03 PDF p10 (Working draft)
S10 PDF p5 (Working draft)

Scope limits

- No cabinet survey observations or installed terminal/channel maps have been supplied for this review.
- No PLC logic, physical interlock or hardware response has been tested for this matrix.
- Full Fermilab accountability for DPS and DPS approval of SC interfaces remain decisions, not recorded agreements.
- Normal operations commands, equipment protection and personnel safety have different owners and paths; a shared display does not merge them.
- No general BMS alarm, network failure or stale REST value is treated as a global detector trip.
- The matrix records source revisions; current/latest status was not revalidated for every cached document.
- The WIEC safety-note download (EDMS 3342150 v2) did not return a PDF on 23 October; its contents were not used to establish a new rule.

Before each stage

These are TTO milestones, not extra automatic trip conditions. The evidence column describes what we need to demonstrate readiness.

ID	Before	Required	Evidence to obtain	Current evidence	Source
R01	Permanent rack power	Rack protection in service for detector-mezzanine, DAQ, CUC and drift racks.	Accepted protection tests, actual AC feeds/load map and final power-off feedback.	Evidence not established in this review	S12 PDF p45, §6.12.3 (TTO requirement)
R02	Top-of-cryostat detector installation	SC rack-status monitoring available.	Tested source-to-display mapping, sensor validity and loss-of-link indication.	Evidence not established in this review	S12 PDF p45, §6.12.3 (TTO requirement)
R03	Beginning detector installation	SC–DPS interface available for the protection needed at that stage.	Agreed scope, executors and function-level tests; resolve D01.	Scope proposal exists; agreement and tests not recorded	S12 PDF pp16,45, §§4.2,6.12.3 (TTO requirement)
R04	Detector components enter underground installation	Bidirectional SC/DPS–SURF Personnel Safety System interface operational.	Owner-approved signal meanings, paths, validity and end-to-end interface tests.	Signal/path agreement not established	S12 PDF p45, §6.12.3 (TTO requirement)
R05	Final cryogenics commissioning / first fill	SC–cryogenics controls interface available.	Agreed point list, process/safety boundaries, quality/latency and test evidence.	Architecture/reference documents exist; accepted interface not established	S12 PDF p45, §6.12.3 (TTO requirement) S14 PDF p1 (Review charge)
R06	TPC/BDE transition to operations	The required electronics safety interlock is operational.	Approved function list and tests against the actual controller/firmware and loads.	Released interface exists; accepted installed functions not established	S12 PDF pp30,33 (TTO requirement) S04 PDF p7 (Stated requirement)
R07	Operations handover	Provide as-builts, channel/cable mapping, installed/spare inventory, procedures, licenses and the software/hardware action matrix.	Reviewed handover package with revisions and accepted test records.	This matrix is an initial draft contribution, not the handover package	S12 PDF p46, §6.12.8 (TTO requirement)

Historical precedent

The 2020 Detector 1 logic is a design reference. It is not an FD-VD setting.

Fan/heater permit: Detector 1 design enables CE LV and APA bias only when Fans and Heat status are good; Heat Only mode disables those supplies while retaining heat. Historical HD/APA logic; not an FD-VD approved rule. S16 PDF p7, §6.3

Fan voting: Fan status is good with X out of 4 fan readbacks running. X is not defined. Do not replace it with an invented voting count. S16 PDF p8, §6.4

Dew-point heating bands: Historical heater command bands are dew point +5/+6 °C for power and +3/+4 °C for low/high selection. These are historical operational heater bands, not approved FD-VD trip limits. S16 PDF p9, §6.5

Sources and revisions

All citations link to online records or pinned public files. Access and approval status differ. Page numbers refer to PDF pages, not printed page labels.

S01 · Slow Control and Detector Safety System Interface Document

EDMS 2907050 v1 · Released · Public EDMS record

Metadata checked 21 October 2026; revision is pinned, not a claim of current approval.

Online record · Exact attachment / pinned file

S02 · Slow Control and Detector Protection System Interface Document

EDMS 2907050 v2 · In Work · Public EDMS record

Metadata checked 21 October 2026; revision is pinned, not a claim of current approval.

Online record · Exact attachment / pinned file

S03 · SC–HVS interface; attachment internal revision v2

EDMS 3315615 v1 · In Work · Public EDMS record

EDMS version 1 contains both v1 and v2 attachments. This matrix uses the v2 PDF, modified 19 October 2026.

Online record · Exact attachment / pinned file

S04 · SC–TPC electronics/BDE interface

EDMS 3293079 v1 · Released · Public EDMS record

Metadata checked 21 October 2026; revision is pinned, not a claim of current approval.

Online record · Exact attachment / pinned file

S05 · TPC electronics/BDE control and monitoring parameter list

EDMS 3362709 v1 · In Work · Public EDMS record

Attachment fetched without sign-in on 23 October 2026. Its alarm bounds are not approved DPS trip limits.

Online record · Exact attachment / pinned file

S06 · PTC Stage 2 Production Readiness Review report

EDMS 3371626 v1 · Released · Public EDMS record

Metadata checked 21 October 2026; revision is pinned, not a claim of current approval.

Online record · Exact attachment / pinned file

S07 · SC–PDS interface; EDMS attachment v1

EDMS 3309688 v1 · In Work · Public EDMS record

Fetched without sign-in on 23 October 2026. This attachment differs from the later public PDS working draft, S08.

Online record · Exact attachment / pinned file

S08 · SC–PDS working draft and proposed interaction matrix

Git commit bfb2a8b725d9a5a3b7a517f4f9e15fb953eed1f8 · Proposed; owner agreement pending · Public GitHub repository

ICD internal revision 0.2; interaction matrix revision 0.1, 17 September 2026. Public file checked 23 October 2026. Not a released replacement for S07.

Online record · Exact attachment / pinned file

S09 · SC–TDE interface; attachment internal revision v2.1

EDMS 3297045 v1 · In Work · Public EDMS record

Fetched without sign-in on 23 October 2026. EDMS version 1 contains older and newer attachments; this matrix uses v2.1.

Online record · Exact attachment / pinned file

S10 · SC–FDC Installation and Integration interface

EDMS 3372107 v1 · In Work · Public EDMS record

Metadata checked 21 October 2026; revision is pinned, not a claim of current approval.

Online record · Exact attachment / pinned file

S11 · SC–DAQ interface

EDMS 3449868 v3.2 · Engineering Check · Public EDMS record

Version 3.2 is under Engineering Check; it is not the released version 3. Source specifies separate normal control and rack/PLC protection paths.

Online record · Exact attachment / pinned file

S12 · Transition to Operations Document for the Far Detector

DUNE DocDB 33801 v1 · Cached document; approval not established · DUNE DocDB sign-in may be required

Cached copy read on 23 October 2026. Latest version not rechecked. Contains software-interlock ownership and PrM/PDS statements that need agreement against current drafts.

Online record · Exact attachment / pinned file

S13 · BSI underground specifications: Section 230923, Instrumentation and Control for HVAC

DUNE DocDB 30544 v2; Bulletin 11, 14 February 2025 · Design specification; installation not checked · DUNE DocDB sign-in may be required

Use PDF pages 399–508 (section pages 230923-1 to -110). Headers identify Bulletin 11. A locally supplied Bulletin 13 exists; its online record and changes have not been established. Do not treat this baseline as current commissioning evidence.

Online record · Exact attachment / pinned file

S14 · LBNF Process Controls Preliminary Design Review charge

EDMS 2386382 v1 · Released · Public EDMS record

2019 review charge, not a completed detector interlock design. Calls for interfaces and failure cases to be reviewed.

Online record · Exact attachment / pinned file

S15 · Far Detector cryogenic controls Final Design Review report

EDMS 2819512 v1 · In Work · Restricted EDMS access

Cached report checked 21 October 2026. Pages 4–5 describe independent process and safety PLC systems with shared supervision. It does not define detector trip limits or prove installation.

Online record · Exact attachment / pinned file

S16 · DDSS Logic Requirements, 28 October 2020

EDMS 2401090 v4 · In Work; historical Detector 1 design · Public EDMS record

Detector 1 APA design. Use only as a historical precedent. Fan voting and dew-point margins are not approved FD-VD settings.

Online record · Exact attachment / pinned file

Source-backed draft for subsystem review. Normal control, protection and personnel safety keep distinct functions. No hardware action has been taken.